

HANDS - ON RAG BOT FOR ECO-TOURISM

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WHAT WE ARE GOING TO DO TODAY?

Steps:

Data Preparation: Load and process the dataset using pandas

Vector Store Setup: Convert product descriptions into embeddings.

Store embeddings in a vector database:

Building the Chatbot:

Use LangChain to create an LLM pipeline.

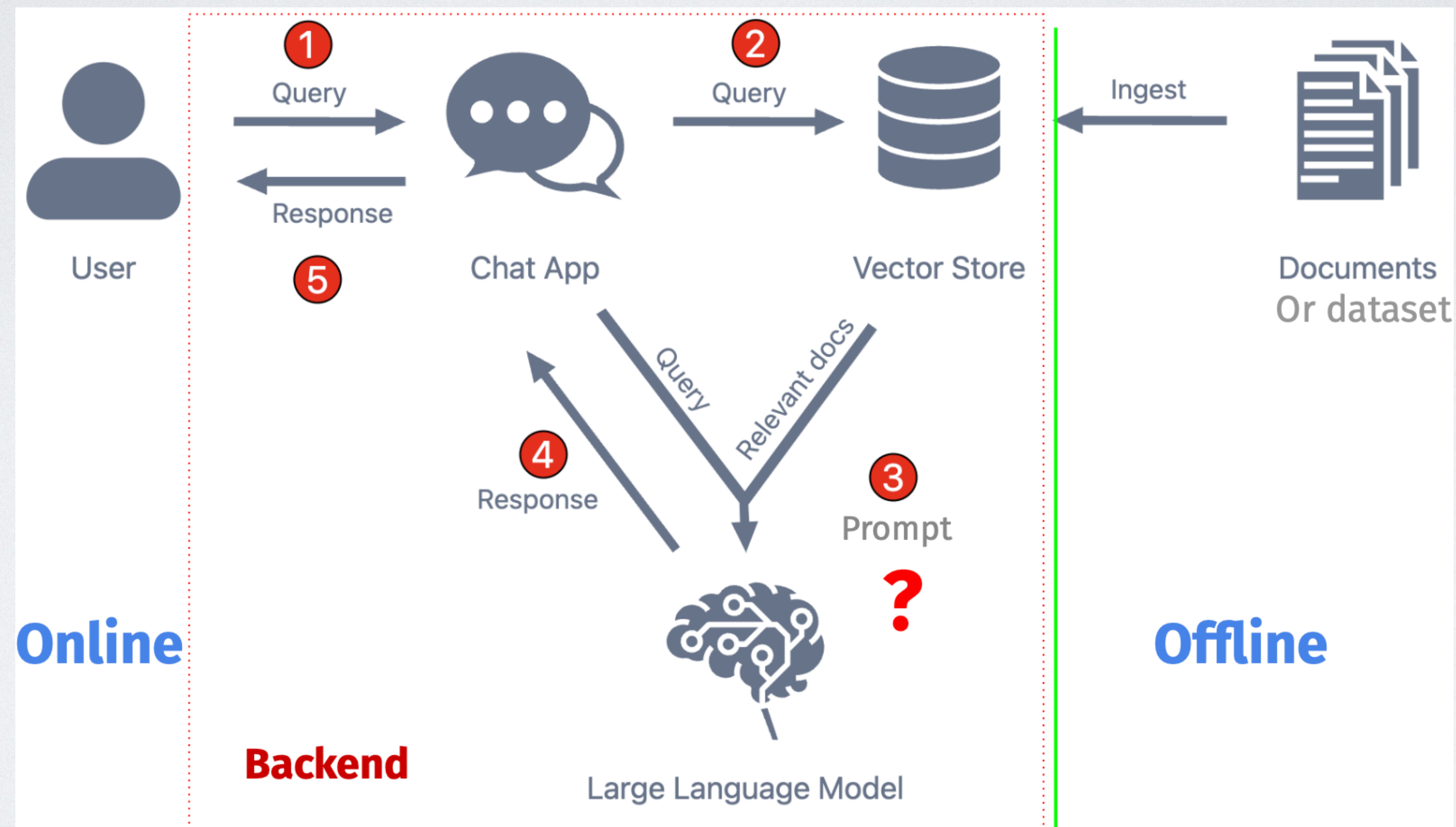
Develop a simple chatbot to answer EcoTourism related queries.

Creating a UI: Implement a Gradio-based UI for user interaction.

WHY RAG IS IMPORTANT

RAG is crucial because LLM models, like ChatGPT, can sometimes "hallucinate" or provide outdated or incorrect information. By retrieving facts from trusted sources before generating responses, RAG ensures the answers are more reliable, up-to-date, and contextually relevant.

RAG ARCHITECTURE



HOW TO BUILD A RAG CHATBOT

- https://github.com/boppanr/Eco_Tourism_Bot
- what is the top location visited
- what is the top season which has more visitor

RAG Evaluation

Metrics:

- a. Helpfulness
- b. Completeness
- c. Correctness

Responsible AI

- a. Harmfulness

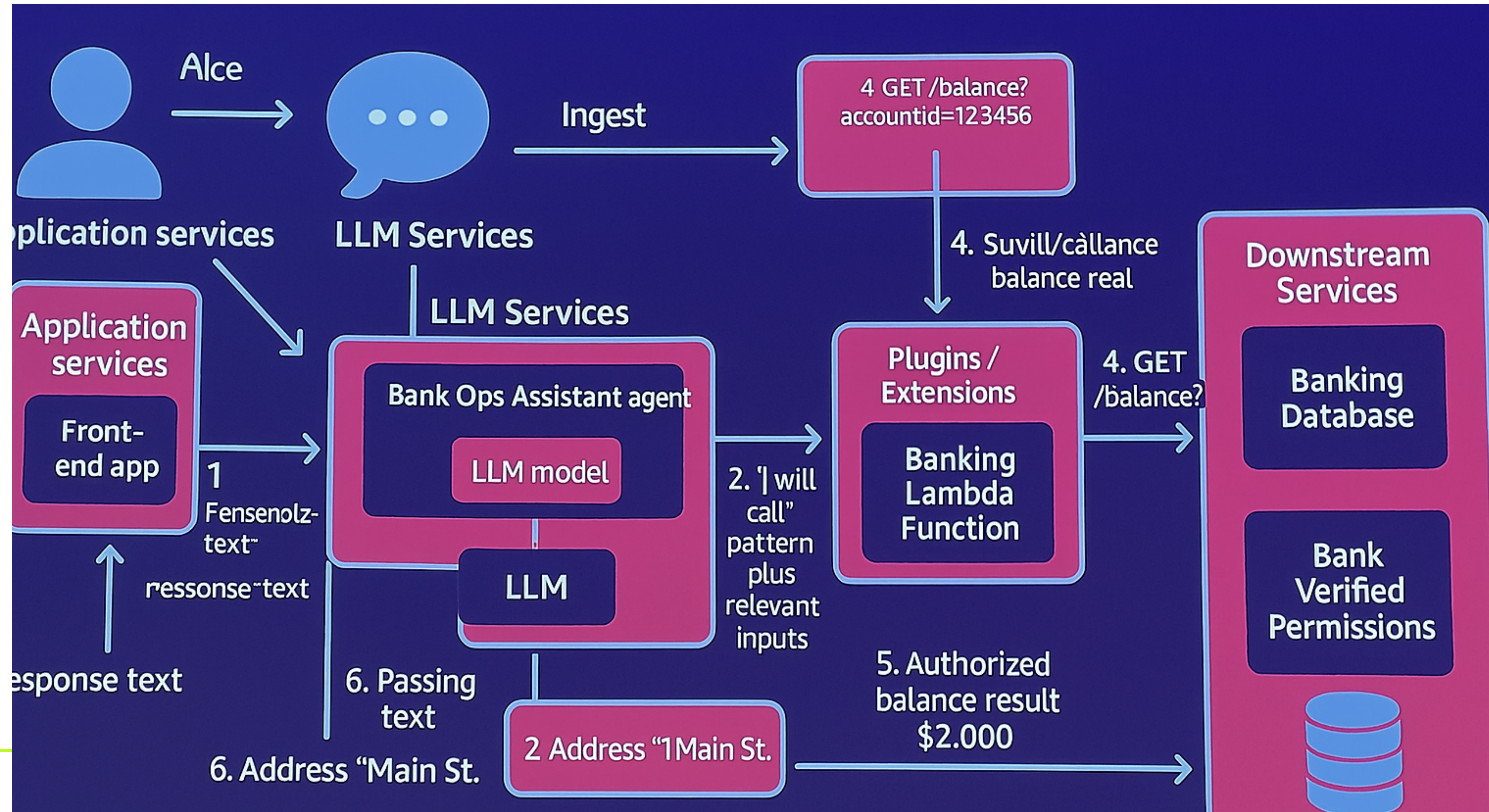
Steps:

1. Setting up the Knowledge Base
2. Uploading & Configuring data
3. Creating Golden Data Set
4. Running evaluations & Metrics
5. Reviewing evaluation Results

HOWTO DEPLOY RAG

- <https://huggingface.co/spaces/rajeshboppana/EcoTourismBot>
- what is the top location visited
- what is the top season which has more visitor

How to Secure RAG/ OWASP Principles



Questions ??
